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Faculty of Computer and Information Technology
Master in Cyber Security

MASTER RESEARCH THESIS

EdgeChain-IDS: A Smart and Decentralized Security Framework for IoT Networks Using Parallel Hybrid IDS, Edge Computing, and Blockchain

Design, Implementation, and Evaluation

Prepared by: Ameen AbdUlmoghni Farhan AL-Hemyari

Supervisor: Dr. Ahmed Al-Shalabi

Program: Master in Cyber Security

Faculty: Faculty of Computer and Information Technology

University: Sana'a University, Yemen

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ABSTRACT

The rapid expansion of the Internet of Things (IoT) into healthcare, industrial automation, smart cities, and enterprise environments has created a security attack surface of extraordinary scale and complexity. IoT ecosystems are characterized by extreme device heterogeneity, severe resource constraints, and latency-sensitive operational requirements, properties that render conventional centralized security architectures fundamentally inadequate. Aggregating detection and response logic in a single cloud-based node creates a high-value target, introduces round-trip latencies that are incompatible with time-critical applications, and, when based on signature matching alone, leaves the network blind to novel and zero-day threats.

This thesis proposes and evaluates EdgeChain-IDS, a smart and decentralized security framework that integrates a Parallel Hybrid Intrusion Detection System (PH-IDS) with Edge Computing and Hyperledger Fabric blockchain. The PH-IDS concurrently operates a signature-based detection engine and an AI-driven anomaly detection engine (Isolation Forest and Long Short-Term Memory networks) at the network edge, and a correlation module applies tiered confirmation thresholds to suppress false positives. Confirmed events are recorded as cryptographically signed transactions on an immutable distributed ledger. Chaincode validates and records the three-tier response policy, while a trusted off-chain response agent applies the corresponding network-enforcement action for high-confidence threats.

Experimental evaluation was conducted in a controlled simulated IoT environment using Scapy and Kali Linux attack toolsets across four attack categories: DDoS, ARP spoofing, botnet activity, and zero-day anomalies. The framework achieved a mean detection accuracy of 96.25% and an aggregate false positive rate of 2.5%, an approximately 57.6% reduction compared with the reported anomaly-engine-alone rate. These figures are bounded by the controlled evaluation and its stated limitations. Edge-based detection latency remained below 35 ms for 500-device networks and below 65 ms at 1000 devices, a greater than 8x improvement over the cloud-based baseline at every tested scale (12x at 500 devices), while automated containment reduced mean response time from 5–30 minutes to under two seconds.

Within the studies included in this thesis' comparative review, no work was identified that simultaneously integrates parallel hybrid intrusion detection at the network edge, blockchain-based immutable security logging, and policy-driven automated response with full empirical evaluation. This bounded observation motivates the proposed framework and should not be read as an exhaustive claim about all published work. The proposed framework is designed to require no modification to existing IoT device firmware and targets commodity edge hardware; deployment claims should be interpreted together with the reported hardware, software, and reproducibility limitations.

Keywords: IoT Security, EdgeChain-IDS, Parallel Hybrid IDS, Edge Computing, Blockchain, Hyperledger Fabric, Anomaly Detection, Isolation Forest, LSTM, Smart Contracts, False Positive Reduction.

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Ameen A. AL-Hemyari
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CHAPTER 1: INTRODUCTION AND RESEARCH FOUNDATION

1.1 Background and Context

The rapid proliferation of Internet of Things (IoT) devices has transformed healthcare, industrial automation, smart cities, and enterprise environments. Connected sensors, actuators, cameras, industrial controllers, and medical monitors now generate and exchange data continuously, enabling levels of automation and situational awareness that were not previously attainable. However, this pervasive connectivity has simultaneously introduced a security attack surface of extraordinary scale and complexity [1, 2].

IoT ecosystems are characterized by three properties that render conventional centralized security architectures fundamentally inadequate. The first is extreme device heterogeneity: a single deployment may include dozens of operating systems, firmware generations, and communication protocols, which makes uniform host-based protection impractical. The second is severe resource constraint: limited CPU, memory, and battery capacity preclude the deployment of conventional security software on the devices themselves. The third is latency-sensitive operation: applications such as industrial control, connected medical devices, and autonomous systems demand security responses within milliseconds rather than minutes.

Intrusion Detection Systems (IDS) are a central component of any network defense strategy. Signature-based IDS compare observed traffic against a database of catalogued attack patterns and offer deterministic, high-accuracy detection with low false positive rates, but they cannot detect novel or zero-day threats. Anomaly-based IDS use machine learning to model normal behavior and flag statistical deviations, which allows them to identify previously unseen attacks, but they generate elevated false positive rates when operated without corroborating evidence [4, 5]. Combining both paradigms is therefore an attractive direction, yet where and how the combined system is deployed has a decisive influence on its operational value.

Edge computing and blockchain technology offer two complementary architectural primitives for addressing these deployment challenges. Edge computing moves analysis to or near the data source and thereby reduces detection latency by an order of magnitude compared with cloud-based models [6]. Blockchain provides decentralized, immutable, and cryptographically verifiable record keeping and, through smart contracts, a mechanism for automated policy enforcement [7, 10]. This thesis investigates whether the convergence of parallel hybrid intrusion detection, edge computing, and blockchain-based automated response can yield a practical and measurably effective self-defending IoT security framework.

1.2 Research Problem

Current IoT security deployments suffer from three specific and interconnected structural vulnerabilities. Each is a consequence of the IoT properties described above and each reduces

the effectiveness of security controls when attacks occur at the speed and scale of modern IoT threats.

First, centralization vulnerability. Aggregating all detection and response logic at a single cloud-based node creates a high-value target. A successful compromise or denial of service against this node disables protection for the entire network simultaneously, and the central node also becomes a single point of failure for log integrity, because logs stored in mutable files can be altered by an attacker who gains privileged access.

Second, latency incompatibility. Routing IoT traffic to remote cloud infrastructure for security analysis introduces round-trip latencies that typically exceed 400 ms. This is operationally incompatible with the sub-100 ms requirements of time-critical IoT applications, for which a delayed detection is functionally equivalent to no detection.

Third, a detection coverage gap. Signature-based IDS are effective against catalogued attack patterns but completely blind to novel, zero-day, and evolving threats, while anomaly-based systems that can detect unknown threats generate unacceptably high false positive rates when deployed in isolation [4, 5]. In addition, response to confirmed threats still depends largely on human intervention, with mean containment times measured in minutes, a window in which fast-spreading threats such as botnet propagation can cause significant damage.

The research problem addressed by this thesis is therefore the absence of an integrated, edge-native security architecture that combines complementary detection paradigms with tamper-proof event logging and automated response, while remaining deployable on resource-constrained infrastructure without modification of IoT devices.

1.3 Scientific Gap

A review of the existing literature shows that substantial progress has been made in several parallel research streams, but that these streams have not been integrated. Early IoT security frameworks established confidentiality, integrity, and availability requirements but overlooked device resource limitations, and later lightweight-cryptography work left real-time threat detection and the latency implications of centralized architectures largely unaddressed [3].

Existing hybrid IDS implementations achieve improved detection rates and reduced false positives [5], yet they remain predominantly centralized in their deployment model, which reintroduces the latency and single-point-of-failure limitations that edge deployment is intended to resolve. Singh et al. [9] explicitly identify the absence of parallel hybrid frameworks operating natively at the network edge as an unresolved open challenge. Edge computing studies treat security as an additional workload rather than the primary design objective [6], while blockchain-based IoT solutions provide immutable logging and policy enforcement but lack an upstream real-time detection component: the immutable log substrate has no mechanism to generate the security events it is intended to record [7, 10]. Finally, the deployment of Isolation Forest and LSTM models on resource-constrained edge devices requires optimization strategies, such as quantization and structured pruning, that are not systematically addressed in most proposed

implementations [8].

The scientific gap, therefore, is the absence of a framework that simultaneously integrates parallel hybrid IDS operation at the network edge, blockchain-based immutable security logging, and smart contract-driven automated response, supported by full empirical evaluation. This thesis addresses that gap through the design and evaluation of EdgeChain-IDS.

1.4 Purpose Statement / Aim

The purpose of this thesis is to design, implement, and evaluate EdgeChain-IDS, a smart and decentralized security framework that integrates a Parallel Hybrid Intrusion Detection System, Edge Computing, and Hyperledger Fabric blockchain to provide rapid, accurate, and tamper-evident detection and automated containment of cyber threats in IoT networks. The framework aims to improve detection accuracy, reduce false positives, minimize detection and response latency, and preserve the integrity of security evidence, all without requiring modification to existing IoT device firmware or software.

1.5 Research Questions

This thesis is guided by the following research questions:

- **RQ1:** What structural limitations of centralized, signature-only, and anomaly-only security approaches reduce their effectiveness in heterogeneous, resource-constrained, latency-sensitive IoT networks?
- **RQ2:** Can a Parallel Hybrid IDS that correlates signature-based and AI-driven anomaly detection engines at the network edge improve detection accuracy and reduce the false positive rate relative to single-paradigm configurations?
- **RQ3:** What detection latency can edge-based processing achieve compared with a cloud-based baseline as the device population grows, and does it satisfy the sub-100 ms requirement of time-critical IoT applications?
- **RQ4:** Can blockchain-based immutable logging combined with smart contract-driven tiered response provide automated containment that is faster than manual intervention while preserving a non-repudiable audit trail?
- **RQ5:** What are the practical deployment requirements, limitations, and scalability considerations of the proposed framework on commodity edge hardware?

1.6 Research Objectives

To address the above research questions, this thesis pursues the following objectives:

- To analyze the structural vulnerabilities of centralized IoT security architectures, namely centralization vulnerability, latency incompatibility, and detection coverage gaps.

- To review and compare existing IoT security frameworks, IDS paradigms, edge computing security, blockchain-based solutions, and AI-driven anomaly detection approaches, identifying their strengths, limitations, and the gap in current knowledge.
- To design a four-layer architecture comprising an IoT Device Layer, an Edge Computing Layer hosting the PH-IDS, a Blockchain Layer based on Hyperledger Fabric, and an AI Inference Layer.
- To develop and optimize Isolation Forest and LSTM models for resource-constrained edge deployment using post-training quantization and structured pruning.
- To implement a smart contract-based three-tier automated response policy with a complete, non-repudiable audit trail.
- To evaluate detection accuracy, false positive rate, detection latency, and automated response performance through controlled experimental attack scenarios, and to derive limitations and directions for future research.

1.7 Significance of the Study

Dimension	Significance
Academic Significance	Contributes a novel edge-native architecture that unifies parallel hybrid intrusion detection, blockchain-based immutable logging, and smart contract-driven response, closing a gap explicitly identified in the literature.
Practical Significance	Provides a deployable security framework on commodity hardware such as Raspberry Pi class gateways, requiring no modification to existing IoT device firmware, software, or communication protocols.
Methodological Significance	Establishes a reproducible quantitative evaluation framework using standard classification metrics (TP, TN, FP, FN) and a systematic edge versus cloud latency comparison across device populations.
Security Significance	Demonstrates measurable improvements in detection accuracy, false positive suppression, and containment speed, and replaces mutable log files with tamper-evident, cryptographically attributable records.
Operational Significance	Reduces dependence on slow manual intervention and alert fatigue by automating containment of high-confidence threats and by providing a non-repudiable audit trail for compliance and forensic investigation.

Table 1: Significance of the Proposed EdgeChain-IDS Framework

1.8 Organization of the Thesis

This thesis is organized into five chapters. Chapter 1 establishes the research foundation, including the background, problem statement, scientific gap, research questions, objectives, and significance. Chapter 2 presents a literature review and theoretical background, synthesizing previous research across IoT security frameworks, intrusion detection, edge computing, blockchain, and AI-driven anomaly detection, and identifying the gap addressed by this work. Chapter 3 describes the research methodology, the four-layer EdgeChain-IDS architecture, the detection and response algorithm, the experimental environment, the attack scenarios, and the evaluation metrics. Chapter 4 reports and discusses the experimental results, including detection accuracy, detection latency, and automated response performance, together with limitations and scalability considerations. Chapter 5 concludes the thesis and outlines directions for future work.

CHAPTER 2: LITERATURE REVIEW AND THEORETICAL BACKGROUND

2.1 Introduction

This chapter reviews the research that is relevant to the design of EdgeChain-IDS and establishes the theoretical background required to understand its components. Research in IoT security has advanced across several parallel streams, each addressing a distinct subset of the broader security problem. The chapter surveys these streams, identifies the contributions and limitations of each, and positions the proposed framework within the resulting landscape.

The chapter is organized as follows. Section 2.2 reviews previous studies on IoT security frameworks, intrusion detection, edge computing, blockchain, and AI-driven anomaly detection. Section 2.3 compares the reviewed approaches and states the research gap. Section 2.4 presents the conceptual framework that guides the thesis, and Section 2.5 provides the theoretical background on the technologies used in the proposed system.

2.2 Review of Previous Studies

2.2.1 IoT Security Frameworks

Early IoT security frameworks established the foundational confidentiality, integrity, and availability (CIA) requirements for connected device ecosystems [2, 3]. Khan et al. [2] described the architecture of the Internet of Things and its key security challenges, while Sicari et al. [3] examined security, privacy, and trust in IoT and outlined the open problems that remained. A consistent limitation of this body of work is that it systematically overlooked the practical constraints imposed by device resource limitations.

Subsequent work incorporated lightweight cryptographic protocols to address this gap. Nevertheless, real-time threat detection capability and the latency implications of centralized architectures remained largely unaddressed. These frameworks define what must be protected, but they do not provide a deployable detection and response mechanism capable of operating at IoT speed.

2.2.2 Intrusion Detection Systems for IoT

IDS approaches for IoT can be categorized into three paradigms. Signature-based IDS offer deterministic, high-accuracy detection for catalogued attack patterns with inherently low false positive rates, but they provide zero detection coverage for novel, evolving, or zero-day threats that are absent from the signature database [4]. Anomaly-based IDS using machine learning overcome this blind spot by detecting statistical deviations from learned normal behavior, which enables identification of previously unseen attack variants, but they suffer from elevated false

positive rates when operated without corroborating evidence [1, 8, 15]. The survey by Al-Garadi et al. [1] reviews machine and deep learning methods for IoT security and highlights both their promise and their deployment difficulties.

Hybrid IDS combining both paradigms achieve improved overall detection rates and reduced false positives [5]. However, existing hybrid implementations remain predominantly centralized in their deployment model, reintroducing the latency and single-point-of-failure limitations that edge deployment is intended to resolve. Singh et al. [9] explicitly identify the absence of parallel hybrid frameworks operating natively at the network edge as an unresolved open challenge in the field. The high false positive rate of anomaly detection is also recognized as the primary factor that erodes the trust of security operations teams in automated detection systems [15].

2.2.3 Edge Computing for Security

Edge computing architectures reduce threat detection latency by an order of magnitude compared with cloud-based models by performing analysis at or near the data source [6], enabling sub-50 ms response for realistically scaled IoT networks. Mao et al. [6] survey mobile edge computing from the communication perspective and describe the latency and bandwidth benefits of local processing. Existing work in this area, however, treats edge computing as a general-purpose computational paradigm and does not co-design security mechanisms as a first-class architectural concern. Security is handled as an additional workload rather than as the primary design objective, which leaves open the question of how detection, logging, and response should be structured when the edge node itself is the primary security enforcement point.

2.2.4 Blockchain for IoT Security

Blockchain-based approaches demonstrate viability for decentralized access control, tamper-proof security event logging, and trustworthy data sharing across heterogeneous IoT device populations [7, 10]. Khan and Salah [7] review IoT security and blockchain solutions and discuss open challenges, and Abadi et al. [10] propose a blockchain-based secure and trustworthy data sharing scheme for IoT.

These solutions provide a strong foundation for security event immutability and for policy enforcement through smart contracts, but they lack an upstream real-time detection component. The immutable log substrate has no mechanism to generate the security events it is intended to record, so its protective value depends on an external detection source that is itself not addressed by these works.

2.2.5 AI-Driven Anomaly Detection

Isolation Forest [11] and LSTM networks [12] are confirmed as effective approaches for high-dimensional IoT traffic outlier detection and temporal sequence anomaly analysis, respectively [8]. A systematic review of AI-based anomaly detection techniques for IoT security [8] reports their effectiveness, but a consistent finding across the literature is that deploying these models on resource-constrained edge devices requires optimization strategies, including quantization

and structured pruning, that are not systematically addressed in most proposed implementations. The N-BaIoT study [5] demonstrates network-based detection of IoT botnet attacks using deep autoencoders and illustrates the value of learned behavioral baselines for IoT traffic.

2.3 Comparative Review and Research Gap

Table 2 presents a comparative analysis of the approaches reviewed in the preceding sections, evaluating each against the six capability dimensions that are relevant to the proposed framework.

Approach	Hybrid IDS	Edge Deployment	Blockchain	Automated Response	Empirical Evaluation
Signature-only IDS	No	No	No	No	Partial
Anomaly-based ML IDS	No	No	No	No	Partial
Edge computing security [6]	Limited	Yes	No	No	Partial
Blockchain IoT [7, 10]	No	No	Yes	No	No
Hybrid IDS [5]	Yes	No	No	No	Partial
EdgeChain-IDS (proposed)	**Yes**	**Yes**	**Yes**	**Yes**	**Full**

Table 2: Comparison of Previous Approaches in IoT Security

The comparative analysis presented in Table 2 reveals a consistent pattern across existing studies: each stream of research addresses one or two of the required capabilities, but none combines all of them. Signature-only and anomaly-based IDS provide detection but neither deployment at the edge nor tamper-proof logging nor automated response. Edge computing security work reduces latency but offers only limited hybrid detection. Blockchain-based work provides immutability but no detection. Hybrid IDS improve accuracy but remain centralized.

Within the studies included in the comparative review, no work was identified that simultaneously integrates parallel hybrid IDS operation at the network edge, blockchain-based immutable security logging, and policy-driven automated response with full empirical evaluation. This bounded observation motivates the design of EdgeChain-IDS; it is not a claim that no such work exists anywhere in the literature.

2.4 Conceptual Framework

The conceptual framework of this thesis is structured around three interconnected dimensions: the threat environment, the proposed architecture, and the evaluation criteria.

The threat environment encompasses the structural vulnerabilities of centralized IoT security (centralization vulnerability, latency incompatibility, and the detection coverage gap) and the

four attack categories used for evaluation: DDoS, ARP spoofing, botnet activity, and zero-day anomalies. These categories span catalogued high-volume attacks, protocol-level manipulation, coordinated multi-stage activity, and novel threats with no signature.

The proposed architecture resolves each vulnerability through a corresponding mechanism. The Parallel Hybrid IDS with a correlation module resolves the detection coverage gap and suppresses false positives. Edge deployment resolves latency incompatibility. Decentralized Hyperledger Fabric peers with an immutable ledger resolve centralization vulnerability and log integrity, and smart contract-driven tiered response resolves the dependence on slow manual intervention.

The evaluation framework measures the performance of the system along four dimensions: detection accuracy (accuracy, true positive rate, and false positive rate under the TP/TN/FP/FN model), detection latency across device populations, automated response performance (containment time and smart contract execution reliability), and log integrity. Evaluating these dimensions together ensures that improvements are assessed holistically rather than in isolation.

2.5 Theoretical Background: Detection, Edge, and Blockchain Technologies

2.5.1 Intrusion Detection Paradigms

A signature-based IDS compares observed traffic against a continuously maintained database of known attack signatures and indicators of compromise. Because detection relies on exact pattern matching, false positive rates are inherently low, but any attack that is not in the database passes undetected. An anomaly-based IDS instead builds a model of normal behavior and flags observations that deviate from it. This can identify previously unseen anomalous behavior, but it does not by itself prove detection of real-world zero-day exploits. Legitimate behavioral variation may also be flagged, which produces false positives.

The two paradigms are complementary: the strength of one corresponds to the weakness of the other. A parallel hybrid design runs both engines concurrently on the same traffic and combines their outputs, so that agreement between independent detectors provides corroborating evidence and a single-engine alert can be handled according to its confidence.

2.5.2 Isolation Forest and LSTM Networks

Isolation Forest [11] is an unsupervised anomaly detection algorithm that exploits the property that anomalous observations, being statistically rare and different from the normal population, can be isolated using fewer random partitions than normal data points. The algorithm constructs an ensemble of isolation trees by recursively partitioning the feature space through random attribute selection and random split-value assignment. The anomaly score of an observation is derived from its average path length across the ensemble: shorter average isolation paths indicate ease of isolation and therefore receive higher anomaly scores.

Long Short-Term Memory (LSTM) networks [12] are a specialized recurrent neural network

architecture incorporating input, forget, and output gates that enable selective retention and forgetting of information across extended temporal sequences. This architecture addresses the vanishing gradient problem that limits standard recurrent networks in learning long-range temporal dependencies, which makes LSTM suitable for modeling normal command-response timing, communication frequency regularity, and protocol sequence conformance of IoT devices.

2.5.3 Edge Deployment and Model Optimization

Edge nodes such as commodity IoT gateways and Raspberry Pi class devices have limited memory and compute capacity compared with cloud servers. Deploying machine learning models on such hardware requires compression. Post-training quantization reduces numerical precision from 32-bit floating point to 8-bit integers, which shrinks model size and accelerates inference, and structured pruning removes low-weight network connections to reduce computational cost. The TensorFlow Lite runtime [14] provides efficient inference within these constraints.

2.5.4 Blockchain, Hyperledger Fabric, and Smart Contracts

Hyperledger Fabric [13] is a permissioned blockchain platform in which participants are authenticated members rather than anonymous nodes. Transactions are validated through a configurable endorsement policy that requires signatures from a specified threshold of peer nodes, are sequenced by an ordering service, and are then committed to a distributed ledger replicated across peers. Once committed, a record is tamper-evident and cryptographically attributable, which provides a non-repudiable audit trail.

Smart contracts in Fabric are implemented as chaincode, which defines the transaction logic that is executed against the ledger. Encoding response policies in chaincode allows policy updates to be version-controlled and deployed without system downtime, and ties each automated response action to an immutable, auditable record.

CHAPTER 3: RESEARCH METHODOLOGY AND SYSTEM DESIGN

3.1 Research Design

This thesis adopts a quantitative design-science methodology. The design-science component governs the construction of EdgeChain-IDS as a deployable, evaluable security artifact. The quantitative experimental component provides the rigorous evaluation framework through which the artifact's effectiveness is measured using standard classification metrics, enabling reproducible comparison with prior work and established baseline configurations.

This approach is appropriate for cybersecurity systems research because it yields reproducible, objective measurements that enable direct comparison with baseline configurations. The design-science paradigm is specifically applicable because the research objective is to create and evaluate a novel artifact that addresses an identified practical problem. The study proceeds through four stages: problem identification and gap analysis (Chapters 1 and 2), artifact design (Section 3.2 and Section 3.3), implementation (Section 3.4), and controlled experimental evaluation (Sections 3.5 to 3.6 and Chapter 4).

3.2 System Architecture and Components

EdgeChain-IDS is structured as a four-layer functional architecture, each layer addressing a distinct dimension of the IoT security challenge, as illustrated in Fig. 1. The architecture is designed to operate without any modification to existing IoT device hardware or firmware, enabling deployment over established device infrastructure.

3.2.1 IoT Device Layer

This layer comprises the heterogeneous population of IoT devices (sensors, actuators, smart appliances, cameras, industrial controllers, and medical monitors) that constitute the primary data sources and attack surfaces of the network. Devices communicate with edge nodes using lightweight, resource-appropriate protocols such as MQTT, CoAP, and constrained HTTP. EdgeChain-IDS imposes no requirements on device firmware, software, or communication protocol, preserving full compatibility with heterogeneous device ecosystems.

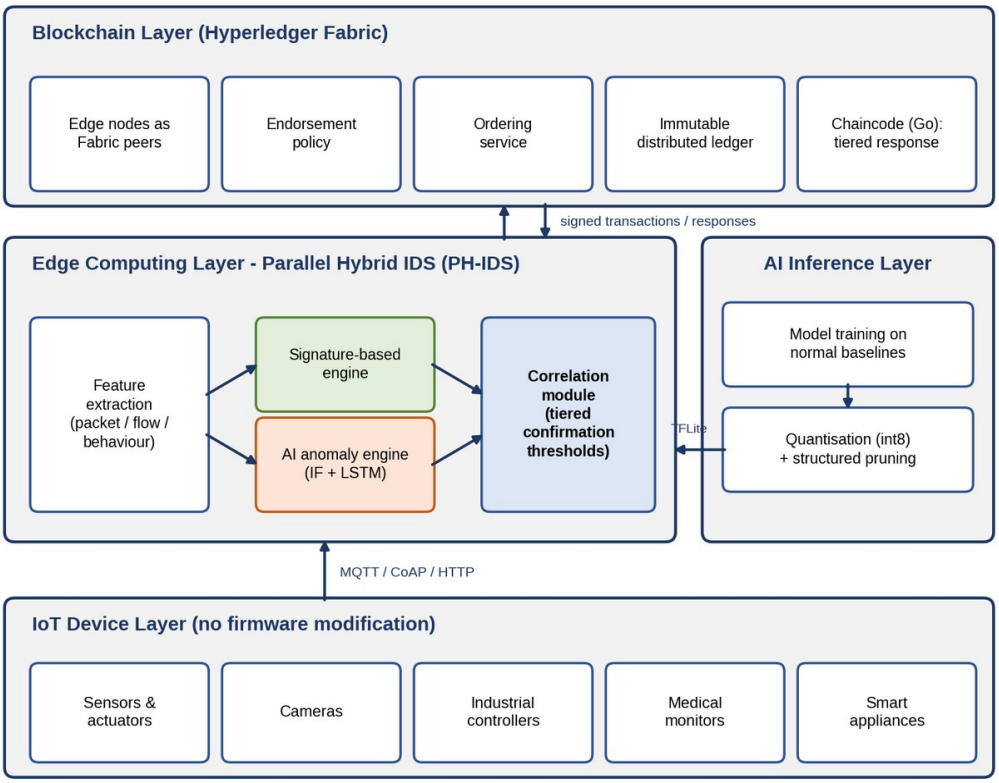


Fig. 1: Proposed EdgeChain-IDS four-layer architecture

3.2.2 Edge Computing Layer and PH-IDS

Edge nodes, which are commodity IoT gateways or local servers such as Raspberry Pi class hardware, serve as the primary security enforcement point. Each edge node hosts the Parallel Hybrid IDS (PH-IDS), which concurrently operates two independent detection engines on the same incoming IoT traffic stream:

- **Signature-based Engine:** Matches incoming traffic against a continuously maintained and updated database of known attack signatures and indicators of compromise. It provides deterministic, high-accuracy detection for all threat patterns present in the database, with inherently low false positive rates arising from exact pattern matching.
- **AI-Driven Anomaly Detection Engine:** Deploys Isolation Forest for statistical outlier detection across high-dimensional traffic feature vectors and LSTM networks for temporal sequence anomaly analysis of device communication patterns. It enables detection of novel, evolving, and zero-day patterns that produce no signature match. The LSTM model is periodically updated through the AI Inference Layer and distributed to edge nodes as an optimized TensorFlow Lite artifact. The Isolation Forest implementation is treated as a separate edge component unless a reproducible conversion and validation procedure is provided.

The outputs of both engines are submitted to a correlation module, which applies tiered confirma-

tion thresholds before triggering automated responses. Only events that satisfy the confirmation threshold, namely dual-engine agreement or single-engine high-confidence detection, trigger smart contract response actions. This correlation mechanism is the primary architectural mechanism for false positive suppression.

3.2.3 Blockchain Layer (Hyperledger Fabric)

The blockchain layer provides a decentralized, immutable, and cryptographically verifiable record of all security events and automated response actions. Edge nodes are configured as Hyperledger Fabric peers, each maintaining a synchronized local copy of the distributed ledger. Upon confirmed threat detection, the edge node generates a cryptographically signed transaction record containing the event timestamp in ISO 8601 UTC format, the affected device identifier (MAC address and IP), the attack type classification, per-engine detection confidence scores, feature attribution data, and the automated response action triggered together with its outcome.

Transactions are validated through Hyperledger Fabric's configurable endorsement policy, which requires endorsement from a specified threshold of peer nodes before submission to the ordering service, and are then committed to the immutable distributed ledger. The resulting record is permanently tamper-evident and cryptographically attributable to specific network events, providing a non-repudiable audit trail for regulatory compliance, forensic investigation, and operational review. Chaincode validates and records response-policy transactions on the ledger. An off-chain event listener or trusted response agent must observe the committed event and invoke the network-enforcement component (such as firewall, SDN, or access-control rules) to perform device isolation. The completed action and its verification result are then recorded as a subsequent ledger transaction.

3.2.4 AI Inference Layer

A centralized server or high-capability edge node hosts the AI model training and continuous update pipeline. The Isolation Forest ensemble and LSTM sequence models are trained on aggregated IoT traffic data representing normal device behavioral baselines across the deployment. Trained models are optimized for edge deployment through post-training quantization and structured pruning, and are distributed to edge nodes via a secure, authenticated update channel for execution on the TensorFlow Lite runtime.

The complete component inventory is summarized in Table 3.

Implementation reproducibility note: the thesis should accompany these components with the actual device specifications, operating-system and library versions, container configuration, model-conversion steps, experiment scripts or source-code location, and raw measurement logs. Without these artifacts, the reported results should be treated as controlled evaluation results rather than independently reproducible benchmarks.

Component Category	Component Details and Function
IoT Device Layer	Heterogeneous sensors, actuators, cameras, industrial controllers, and medical monitors communicating over MQTT, CoAP, and constrained HTTP. No firmware modification required.
Edge Nodes (PH-IDS)	Commodity gateways (e.g., Raspberry Pi 4, 4 GB RAM) hosting containerized signature and anomaly engines, a feature extraction pipeline, and the correlation module.
Blockchain Layer	Hyperledger Fabric single-channel permissioned network with edge nodes as peers, endorsement policy, ordering service, immutable ledger, and Go chaincode.
AI Inference Layer	Model training and update pipeline; post-training quantization (float32 to int8) and structured pruning; secure model distribution as TensorFlow Lite artifacts.
Automated Response	Smart contract-driven three-tier response policy (Critical, High, Medium) with outcomes recorded as subsequent ledger transactions.

Table 3: Components of the proposed EdgeChain-IDS framework

3.3 Detection and Response Algorithm

The proposed system implements a multi-stage detection and automated response algorithm that operates continuously at each edge node. The workflow of the algorithm is illustrated in Fig. 2, and its stages are as follows:

- **Stage 1: Continuous Monitoring.** The edge node receives IoT traffic from devices over lightweight protocols and monitors the incoming stream continuously.
- **Stage 2: Feature Extraction.** Traffic undergoes a preprocessing pipeline that extracts packet-level, flow-level, and behavioral features, producing a combined multi-dimensional feature vector.
- **Stage 3: Parallel Detection.** The same feature vector is presented simultaneously to the signature-based engine and the AI anomaly engine, which process it independently.
- **Stage 4: Correlation and Confirmation.** The correlation module combines the per-engine confidence scores and applies tiered confirmation thresholds. Events that do not satisfy the threshold are logged and monitoring continues.
- **Stage 5: Tier Assignment.** Confirmed events are assigned a response tier (Critical, High, or Medium) according to the detection confidence of both engines.
- **Stage 6: Blockchain Consensus and Logging.** A cryptographically signed transaction is

generated, endorsed by the configured threshold of peers, ordered, and committed to the immutable ledger.

- **Stage 7: Automated Response.** After the policy transaction is committed, an off-chain trusted response agent receives the event and invokes the network-enforcement component to apply the tiered action. Chaincode records the policy decision; it does not directly modify firewall or network rules.
- **Stage 8: Feedback and Audit.** The response outcome is recorded as a subsequent blockchain transaction, completing the chain of custody and supporting continuous monitoring and model performance review.

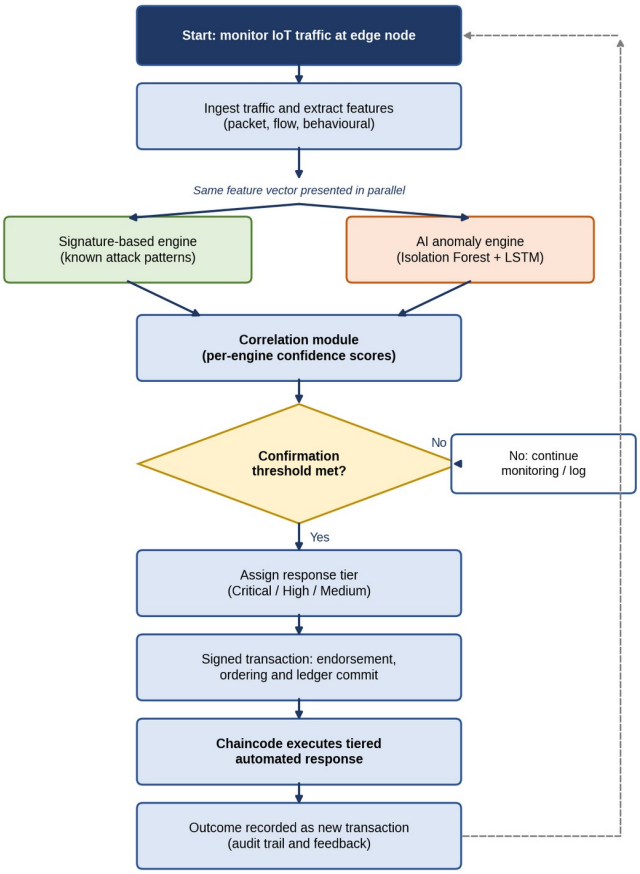


Fig. 2: Workflow of the proposed detection and automated response algorithm

The response policy is organized into three tiers, summarized in Table 4. Tier assignment is determined by the correlation module based on detection confidence levels from both engines.

Tier	Trigger Condition	Automated Response Actions
Critical	Dual-engine high-confidence confirmation	Immediate device isolation blocking all traffic from and to the device; access credential revocation; administrator alert with the full event record; enhanced monitoring of network-adjacent devices.
High	Single-engine elevated-confidence detection	Traffic rate limiting on the affected device; enhanced logging at five times the normal granularity; administrator notification pending second-engine confirmation.
Medium	Multiple medium-confidence events from the same source	Behavioral flagging in the security event database; increased monitoring polling cadence; retrospective analysis queue entry for human review.

Table 4: Smart Contract Response Tiers

All automated response actions and their outcomes are recorded as subsequent blockchain transactions, maintaining a complete, non-repudiable chain of custody from initial detection through response execution to outcome verification. Response policies are encoded in Hyperledger Fabric chaincode written in Go, which enables version-controlled policy updates without system downtime.

3.4 Experimental Environment and Tools

EdgeChain-IDS was evaluated in a controlled simulated IoT network environment incorporating realistic heterogeneous device traffic profiles. Standard IoT gateways or commodity single-board computers (for example, a Raspberry Pi 4 with 4 GB RAM and an ARM Cortex-A72 processor) serve as the target edge node class. Each node runs a lightweight Linux-based operating system and hosts containerized PH-IDS modules using Docker for isolation and deployment management. Table 5 lists the tools and technologies used.

Tool / Technology	Purpose
Scapy and Kali Linux	Generation of attack traffic (DDoS, ARP spoofing, botnet activity, zero-day anomalies).
Raspberry Pi 4 class node	Target edge hardware (4 GB RAM, ARM Cortex-A72) with a 512 MB RAM model deployment profile.
Docker	Containerized PH-IDS modules for isolation and deployment management.
TensorFlow Lite	Efficient inference of optimized Isolation Forest and LSTM artifacts on edge hardware.
Hyperledger Fabric v2.5	Permissioned blockchain, endorsement policy, ordering service, and immutable ledger.
Go chaincode	Encoding of the tiered automated response policies.
MQTT, CoAP, constrained HTTP	Lightweight device-to-edge communication protocols.

Table 5: Tools and Technologies Used in the Proposed System

3.4.1 Feature Extraction

IoT traffic received at each edge node undergoes a preprocessing and feature extraction pipeline before it is fed simultaneously to both detection engines. Features are extracted across three levels of abstraction:

- **Packet-level features:** source and destination IP address and port, packet size, inter-arrival time, protocol type, and TCP flag combinations.
- **Flow-level features:** flow duration, total packet and byte counts, packets and bytes per second, forward-to-backward directionality ratio, and flow idle time.
- **Behavioral features:** device communication frequency, command-response timing intervals, protocol sequence conformance scores, and temporal communication regularity.

3.4.2 Model Configuration and Optimization

The Isolation Forest ensemble comprises 100 isolation trees, with the contamination parameter set empirically based on the expected anomaly prevalence of the deployment environment. The LSTM model is a two-layer stacked architecture with 128 units in the first layer and 64 units in the second, dropout regularization at a rate of 0.2, and input sequences of 50 timesteps. It is trained on IoT device communication event sequences to learn normal command-response timing, communication frequency regularity, and protocol sequence conformance, and it flags statistically significant temporal deviations as anomaly indicators.

Post-training quantization reduces model weights from float32 to int8 precision, achieving approximately 75% model size reduction with less than 2% accuracy degradation in validation testing. Structured pruning removes network connections with weights below a configured significance threshold, further reducing computational cost without meaningful accuracy loss. These optimizations enable deployment within the 512 MB RAM profile of the target edge hardware class.

3.4.3 Blockchain Configuration

Hyperledger Fabric is configured as a single-channel permissioned network with edge nodes as authorized peer participants. Each confirmed threat detection event generates a transaction that is submitted to the ordering service. Endorsement requires signatures from a configurable threshold of peer nodes before the ordering service commits the transaction to the immutable ledger, which ensures decentralized validation and tamper-evident logging.

3.5 Attack Scenarios

Attack traffic was generated using Scapy and Kali Linux toolsets across four representative attack categories. Each category was evaluated across 100 test events comprising 50 attack instances and 50 legitimate traffic instances, which enables statistically meaningful TP/TN/FP/FN classification analysis.

3.5.1 DDoS Attack Scenario

High-volume UDP and ICMP flooding was directed at simulated IoT devices to exhaust their resources and disrupt service. This is a catalogued attack with a distinctive flow-level volume signature, and it is expected to be detected through signature matching with anomaly engine corroboration.

3.5.2 ARP Spoofing Attack Scenario

Forged ARP reply packets were injected to poison the ARP caches of target devices and redirect traffic through an attacker-controlled host. The attack manipulates the trust-based address resolution mechanism and exhibits a highly distinctive statistical signature in flow-level features.

3.5.3 Botnet Activity Scenario

Botnet behavior was emulated through command-and-control (C2) server beaconing and lateral movement patterns. These coordinated, multi-stage behaviors represent the propagation threat for which rapid automated containment is most valuable.

3.5.4 Zero-Day Anomaly Scenario

Novel protocol fuzzing and behavioral deviations that were not present in the signature database were generated as representative unseen anomalies. These events test anomaly detection, but they do not by themselves establish detection of real-world zero-day exploits. Their interpretation depends on documenting the training exclusion, the train/test separation, and the generation procedure.

3.6 Evaluation Metrics

The effectiveness of EdgeChain-IDS was evaluated using the following standardized performance metrics, selected to provide a multidimensional assessment of detection quality, speed, and operational impact.

Metric	Description	Measurement Unit
Detection Accuracy	Proportion of all test events (attack and legitimate) that are correctly classified	Percentage (%)
True Positive Rate (Recall)	Proportion of attack events that are correctly detected	Percentage (%)
False Positive Rate	Proportion of legitimate events incorrectly classified as attacks	Percentage (%)
Detection Latency	Time required by the system to detect and classify a threat event	Milliseconds (ms)
Containment Time	Time from confirmed detection to completed containment action	Seconds / minutes
Smart Contract Execution Rate	Proportion of confirmed alerts for which the response was executed successfully	Percentage (%)

Table 6: Evaluation Metrics and Descriptions

Detection performance was computed using the standard classification model equations:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{TPR (Recall)} = \frac{TP}{TP + FN}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

where TP (True Positive) represents correctly detected attacks, TN (True Negative) represents correctly identified legitimate traffic, FP (False Positive) represents legitimate activity incorrectly

classified as an attack, and FN (False Negative) represents undetected malicious activity. Latency was compared against a cloud-based baseline across device populations of 100, 250, 500, 750, and 1000 devices, and automated response was compared against a human-in-the-loop manual response baseline.

3.7 Operational Workflow

The complete data flow of the system proceeds through six sequential stages, representing its behavior from data ingestion to audit:

- **Step 1: Data Ingestion.** IoT devices transmit operational data to edge nodes via lightweight secure protocols.
- **Step 2: Edge Processing.** The PH-IDS performs feature extraction and parallel detection engine analysis in real time.
- **Step 3: Threat Alerting.** Confirmed threat events generate cryptographically signed alert transactions.
- **Step 4: Blockchain Consensus and Logging.** Transactions are validated through the endorsement policy, ordered, and committed to the immutable ledger.
- **Step 5: Automated Response.** Triggered smart contracts execute the appropriate tiered response action.
- **Step 6: Feedback and Audit.** Response outcomes are recorded as blockchain transactions, creating a complete audit trail and enabling continuous monitoring and model performance review.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Overview of Experimental Results

EdgeChain-IDS was evaluated in a controlled simulated IoT network environment incorporating realistic heterogeneous device traffic profiles. Four attack categories were tested: DDoS, ARP spoofing, botnet activity, and zero-day anomalies. Each category comprised 100 test events (50 attack instances and 50 legitimate traffic instances), and the results were analyzed along three dimensions: detection accuracy, detection latency, and automated response performance.

The following sections present the detailed results for each dimension, followed by a discussion of the findings in relation to the research questions, the limitations of the study, and the scalability of the framework.

4.2 Detection Accuracy Results

Table 7 presents the complete classification results for the correlated dual-engine PH-IDS across the four attack categories.

Attack Category	TP	TN	FP	FN	Accuracy	TPR (Recall)	FPR
DDoS	48	49	1	2	97.0%	96.0%	2.0%
ARP Spoofing	49	49	1	1	98.0%	98.0%	2.0%
Botnet Activity	47	49	1	3	96.0%	94.0%	2.0%
Zero-Day Anomaly	46	48	2	4	94.0%	92.0%	4.0%
Average	**47.5**	**48.75**	**1.25**	**2.5**	**96.25%**	**95.0%**	**2.5%**

Table 7: Detection Accuracy per Attack Category ($n = 100$ test events per category: 50 attack, 50 legitimate)

ARP spoofing achieved the highest accuracy (98.0%) owing to the highly distinctive statistical signature of forged ARP reply traffic in flow-level features. DDoS detection reached 97.0% and botnet activity 96.0%. Zero-day anomaly detection reached 94.0%. This category relies entirely on the anomaly detection engine, since these events by definition produce no match in the signature database; the signature engine contributes zero detection coverage for this category, which confirms the essential and non-substitutable role of the AI-driven component.

The four false negatives recorded in the zero-day category reflect edge cases in which novel attack patterns exhibited insufficient statistical deviation from the 50-timestep LSTM evaluation baseline within the assessment window. This identifies extended temporal context modeling as a

primary direction for future improvement.

4.3 Comparison with Baseline Configurations

The correlated dual-engine PH-IDS architecture achieves a mean detection accuracy of 96.25% across all four attack categories, outperforming a traditional single-paradigm IDS baseline (86.7%) by 9.55 percentage points and the anomaly engine operating in isolation (93.0%) by 3.25 percentage points. These differences are calculated from the accuracy values reported in Table 7 and should not be interpreted as evidence beyond the tested sample. Table 8 summarizes the comparison.

Configuration	Detection Accuracy	False Positive Rate
Traditional single-paradigm IDS (baseline)	86.7%	Not reported
Anomaly engine operating alone	93.0%	5.9%
EdgeChain-IDS (correlated dual-engine PH-IDS)	**96.25%**	**2.5%**

Table 8: Detection Performance of EdgeChain-IDS versus Baseline Configurations

The aggregate false positive rate calculated from the TP/TN/FP/FN counts in Table 7 is 2.5%. Relative to the anomaly-engine-alone configuration (5.9%), this represents an approximately 57.6% reduction. This attribution is limited to the controlled evaluation and should be interpreted alongside the stated sample-size and measurement limitations. Because only events that satisfy the confirmation threshold trigger automated response, independent corroboration by the signature engine suppresses anomaly alerts that have no supporting evidence.

4.4 Attack Detection by Category

DDoS, ARP spoofing, and botnet events were detected through dual-engine corroboration in all confirmed cases, with the signature engine providing the primary detection signal and the anomaly engine providing independent corroboration that enables the correlation module to suppress false positive alerts. Zero-day anomalies and novel evolving attack patterns were identified exclusively through the anomaly detection engine, as summarized in Table 9.

Attack Category	Primary Detection Signal	Corroboration	Accuracy
DDoS	Signature engine	Anomaly engine (dual-engine)	97.0%
ARP Spoofing	Signature engine	Anomaly engine (dual-engine)	98.0%
Botnet Activity	Signature engine	Anomaly engine (dual-engine)	96.0%
Zero-Day Anomaly	Anomaly engine only	None (no signature exists)	94.0%

Table 9: Detection Path by Attack Category

This distribution indicates complementary coverage between the two detection engines. Based on the counts in Table 7, the correlated architecture achieved 96.25% mean accuracy and a 2.5% aggregate false positive rate in the reported evaluation.

4.5 Detection Latency

Edge-based processing maintains significantly lower detection latency than cloud-based alternatives across all tested device population sizes. Table 10 and Fig. 3 present the systematic comparison.

Measurement boundary: the latency values in Table 10 are detection-and-classification latency as reported, not necessarily end-to-end containment latency. The measurement protocol should state whether packet transmission, feature extraction, blockchain endorsement/commitment, network isolation, and response verification are included, and should report the number of repetitions and the summary statistic used.

Device Population	EdgeChain-IDS Edge Latency	Cloud-Based Baseline Latency	Latency Improvement Factor
100 devices	∼15 ms	∼200 ms	13.3x
250 devices	∼22 ms	∼310 ms	14.1x
500 devices	∼35 ms	∼420 ms	12.0x
750 devices	∼48 ms	>460 ms	>9.6x
1000 devices	∼62 ms	>510 ms	>8.2x

Table 10: Detection Latency: EdgeChain-IDS Edge Processing versus Cloud-Based Baseline

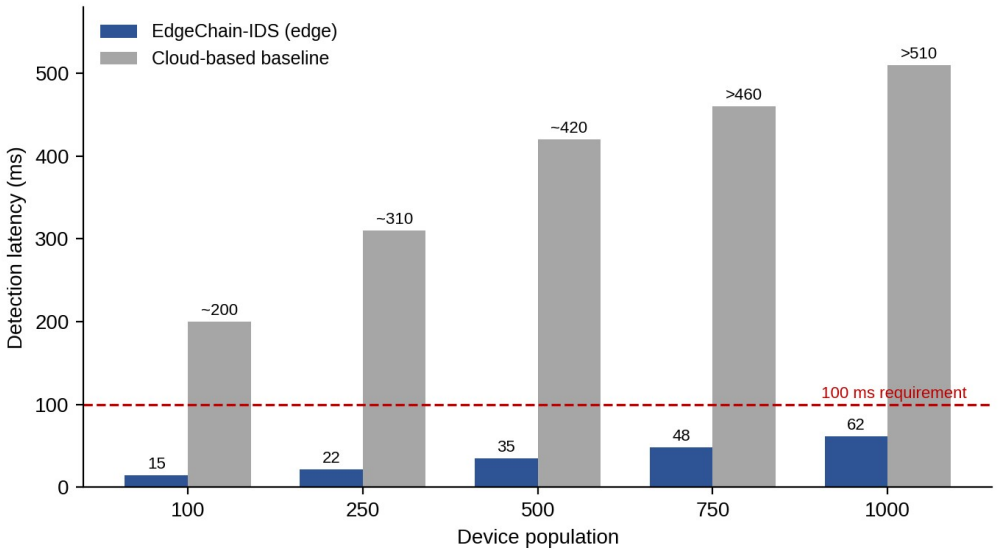


Fig. 3: Detection latency of EdgeChain-IDS versus the cloud-based baseline (values marked $>$ are lower bounds)

EdgeChain-IDS maintains sub-65 ms latency at 1000 devices, satisfying the sub-100 ms operational requirement of industrial IoT and connected medical device applications. The cloud-based baseline fails to satisfy this requirement at any tested population size, exceeding 510 ms at 1000 devices. At 500 devices, edge latency is approximately 35 ms against approximately 420 ms for the cloud baseline, a 12-fold improvement.

4.6 Automated Response Performance

Smart contract-driven automated response reduced mean time to containment from 5–30 minutes (human-in-the-loop manual response baseline) to under 2 seconds for Critical-tier confirmed threats. Smart contract execution succeeded for 100% of confirmed Critical-tier alert events across all four attack categories. Table 11 summarizes the response performance comparison.

Evidence boundary: the reported 100% execution rate is meaningful only with the denominator, success definition, test repetitions, and operating conditions. In particular, successful chaincode execution should be distinguished from verified enforcement of the external isolation action.

Performance Metric	Manual Baseline	EdgeChain-IDS Automated
Mean detection time	$>$5 minutes	12–35 ms
Mean containment time	5–30 minutes	$<$2 seconds
False positive rate	$\sim$12% (signature-only)	2.5% (correlated)
Security log integrity	Mutable log files	Immutable blockchain
Smart contract execution rate	N/A	100% of confirmed alerts

Table 11: Automated Response Performance: Manual Baseline versus EdgeChain-IDS

Beyond speed, the blockchain layer changes the evidentiary quality of the security record. Each detection and each response action is stored as a signed, endorsed transaction, so the audit trail cannot be silently altered after an incident, in contrast with mutable log files.

4.7 Discussion

The experimental results confirm the three primary technical claims of EdgeChain-IDS, which correspond to research questions RQ2 to RQ4.

Parallel hybrid detection superiority (RQ2). The 96.25% correlated accuracy versus 93.0% for the anomaly engine alone and 86.7% for the single-paradigm baseline, combined with the approximately 57.6% false positive rate reduction calculated from the reported counts, indicates that dual-engine correlation was effective in this controlled evaluation. Generalization requires larger, independently separated test sets. The false positive reduction is particularly significant for operational deployment, because high false positive rates are the primary factor cited by security operations teams for the erosion of trust in automated detection systems [15].

Edge deployment operational necessity (RQ3). The consistent sub-65 ms edge detection latency versus cloud-based latencies exceeding 510 ms represents not merely a quantitative improvement but an operational threshold: for industrial control systems requiring sub-100 ms security response, the cloud-based architecture is not a slower alternative but a non-viable one. The latency improvement factor remains above 8x across all tested population sizes (100 to 1000 devices), which indicates that the edge latency advantage is robust to deployment scale.

Smart contract automation effectiveness (RQ4). The reported reduction in mean containment time—from a minimum of 5 minutes to under 2 seconds—and the reported 100% execution rate for confirmed Critical-tier threats suggest that automation can reduce the human-in-the-loop latency gap. However, the thesis must report the number of attempts, the success definition, and whether successful network isolation was verified separately from transaction execution before this result can be generalized.

4.8 Limitations

Several limitations should be considered when interpreting these results (RQ5):

- **Simulation environment.** The simulation, while incorporating realistic device traffic profiles, cannot fully replicate the hardware variability, firmware heterogeneity, and operational complexity of live production deployments. Performance validation in live enterprise environments is a necessary next step.
- **Zero-day detection boundary.** The four zero-day false negatives reflect the fundamental detection boundary of anomaly-based systems: novel patterns exhibiting insufficient deviation from the learned baseline within the 50-timestep evaluation window will not be detected. Extended temporal context modeling using attention mechanisms or longer LSTM sequences is the primary identified mitigation.

- **Blockchain overhead.** Computational and storage overhead of the blockchain layer under sustained high-volume attack conditions at device populations exceeding 1000 requires further quantification before production deployment decisions can be made with full confidence.
- **Model dependence on training data.** AI model detection performance is contingent on training data representativeness and may require periodic retraining as IoT device populations and traffic patterns evolve.

4.9 Scalability

The modular layered architecture supports deployment scale through hierarchical edge node organization, configurable Hyperledger Fabric consensus parameters, and federated learning strategies for distributed AI model updates that avoid centralized aggregation of raw traffic data. The consistent latency improvement factor across all five tested device population sizes indicates that the edge computational advantage scales predictably with network growth.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This thesis has presented the design, implementation, and experimental evaluation of EdgeChain-IDS, a smart and decentralized security framework for IoT networks designed to resolve three interconnected structural limitations of existing security approaches: centralization vulnerability, latency incompatibility, and detection coverage gaps. The framework integrates a Parallel Hybrid IDS operating at the network edge, Hyperledger Fabric blockchain-based immutable security event logging, and smart contract-driven automated tiered threat response into a unified, self-defending architecture that is deployable without modification to existing IoT device infrastructure.

The research addressed a bounded gap identified through the comparative review in this thesis: within the reviewed studies, no work was identified that simultaneously integrates parallel hybrid intrusion detection at the network edge, blockchain-based immutable logging, and policy-driven automated response with full empirical evaluation.

Experimental evaluation across four representative IoT attack categories demonstrated a mean detection accuracy of 96.25%, exceeding the reported single-paradigm IDS baseline by 9.55 percentage points; an aggregate false positive rate of 2.5%, representing an approximately 57.6% reduction versus the reported anomaly-engine-alone rate; sub-65 ms edge-based detection latency at 1000 devices, an improvement of more than 8x over the reported cloud-based baselines; and reported automated containment in under 2 seconds for high-confidence threats. These results are bounded by the controlled test environment and the sample and measurement limitations discussed in this thesis.

In conclusion, this research presents evidence that the convergence of parallel hybrid intrusion detection, edge computing, blockchain immutability, and policy-driven automation is technically plausible and was evaluated in a controlled environment. Claims of deployment on commodity hardware should be read together with the implementation and reproducibility details reported in this thesis. The framework offers a practical path toward IoT networks that defend themselves at the speed at which they are attacked, while preserving a trustworthy record of every security event.

5.2 Future Work

The findings of this research suggest several promising directions for future investigation and system enhancement:

- **Federated Learning:** Privacy-preserving distributed model updates would allow edge nodes to improve shared models without centralizing raw traffic data.

- **Live Enterprise Deployment Validation:** Deployment and evaluation in live environments under real-world hardware and operational conditions are required to confirm the simulation findings.
- **Extended Temporal Context:** LSTM sequences of 100 to 500 timesteps and attention mechanisms should be investigated to reduce zero-day false negative rates.
- **Formal Verification of Smart Contracts:** Formal verification of the chaincode security logic would strengthen assurance in the automated response policies.
- **Multi-Edge Cooperative Detection:** Cooperation between edge nodes would enable network-wide threat correlation.
- **Energy Consumption Profiling:** Energy profiling is needed for battery-powered and energy-harvesting IoT gateway deployments.

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